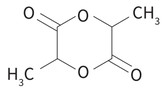
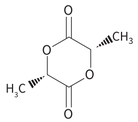
AestheFill® Key Component PDLLA (Poly-D, L-Lactic Acid)

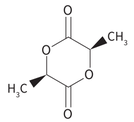


Lactide

$C_6H_8O_4$



L-Lactide



D-Lactide

PDLLA (Poly-D, L-Lactic Acid) is an aliphatic thermoplastic polymer derived from 100% renewable resources, such as corn and potato starch. It is widely used in the medical industry (injectables, bio-degradable dental implants & slow-release drugs) with high safety.

Lactic acid has two optical isomers, L-lactic acid and D-lactic acid, with the same chemical and physical properties. Through polymerization, PDLLA is formed.

Optical Isomers

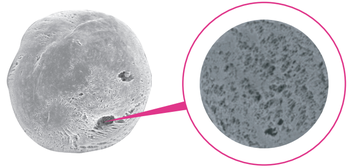
Optical Isomers: have the same molecular formula, but have a different arrangement of the atoms.

AestheFill® Structure

Porous Microsphere

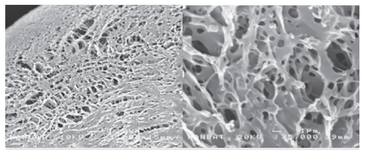
Surface Structure

High Density

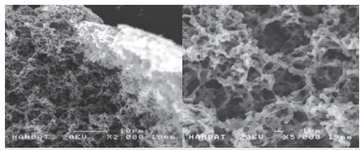


Inner Structure

Empty



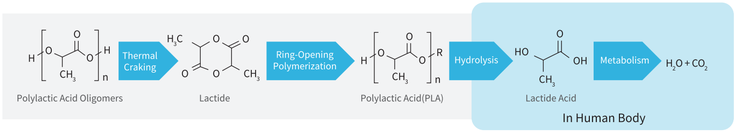
Surface Structure



Inner Structure

Safety of AestheFill®

1. Biodegradable: Through hydrolysis, PDLLA is decomposed into CO_2 and H_2O, and is excreted outside the body.
2. There is no migration of PDLLA into other tissue after injection.



AestheFill® Q&A

What is AestheFill® ?

AestheFill is a filler which enables the human body to stimulate their own collagen and the results can last from 1.5 to 2 years. The main material is PDLLA which have been proven it is safe to be used in the human body as our body has the ability to decompose and metabolize the material. The stimulation of collagen and filling in the concave area on the skin enables the skin to look naturally plump and smooth.

How does the effect change overtime?

AestheFill uses a porous microsphere as a base for the stimulation of collagen. The collagen fills up the wrinkles, fine lines, and any concave parts of the face, giving a more youthful and refined appearance. The result can be seen gradually and subtly over time and last over a period of 24 months.

How many sessions are required?

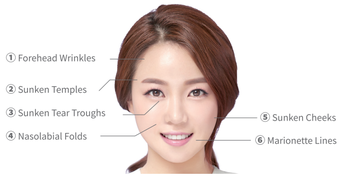
To obtain the best results, a complete treatment of 1~3 sessions are recommended with a time interval of 4~6 weeks between each treatment. However, depending on the person, the number of treatments and time will vary. Please seek a professional for further planning.

How long does it take after the treatment to see the results? How long can the result last?

Results can be seen after the moisture is absorbed. Depending on the concentration injected; you can see the effect after around 1 week if it's a high concentration. For normal concentration, the results will gradually occur and you'll be able to see improvements in fine lines/wrinkle and more refined appearances. Base on different people's physiques and living habits, the ability to stimulate collagen will be different; the results and how long it'll last depends on the person.

Who is suitable for the use of AestheFill?

AestheFill can be used on the entire face, including the temples, apples cheeks, and any skin cavities. Also, the nasolabial fold, marionette line, contour lines, the loosening and sagging of the skin can be improved by the treatment of AestheFill. It is also suitable to improve the fine lines on your hand and neck.

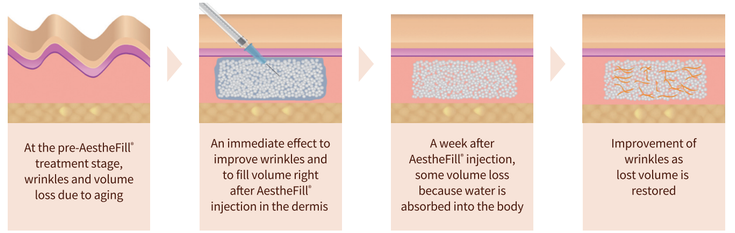


What are the possible side effects after injection?

The key component of AestheFill®, PDLLA is an FDA approved material derived from 100% natural botanic resources. Although the treatment is very safe, some treatment-related reactions may occur. Slight redness and bruising will only last for a few hours to a week. Hematoma may also happen and would disappear in a few thours to two weeks. Other small papules under the skin in the treated area may happen. These small papules may not be visible, and you may notice them only when you press on the treated skin.

AestheFill® Mechanism

1. Immediate volume-filling effect after injection.



At the pre-AestheFill® treatment stage, wrinkles and volume loss due to aging

An immediate effect to improve wrinkles and to fill volume right after AestheFill® injection in the dermis

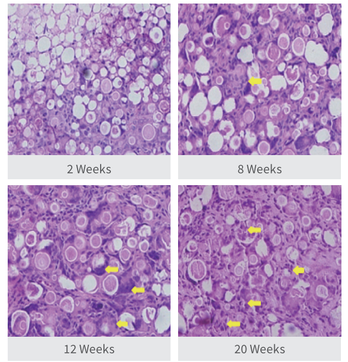
A week after AestheFill® injection, some volume loss because water is absorbed into the body

Improvement of wrinkles as lost volume is restored

2. Bio-stimulation: Stimulation of new tissues and collagen production.

Monocyte Migration at the injected area

Production of new tissues confirmed by histological analysis.



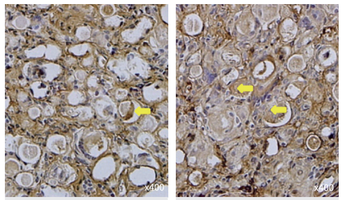
2 Weeks | 8 Weeks | 12 Weeks | 20 Weeks

Pink: Cytoplasm Dark pink or Blue: Nucleus Arrows: Foreign body giant cells

- At Week 2 after injection, cells have grown and attached on the surface of PDLLA particles and to spaces between particles. The empty spaces between particles have been filled with cells.
- At Week 8, there are no empty spaces between PDLLA particles, by being filled with cells which start to migrate into porous PDLLA particles (yellow arrow).
- At Week 12 and 20, cells have migrated into the inner parts of PDLLA particles, and fill inside and spaces between the particles.

Histological analysis - IHC staining for actin and Type1 collagen

Production of new tissues confirmed by histological analysis.



After 12 Week | After 20 Week

- From Week 4, new collagen has been generated by the cells settled in the PDLLA.
- At Week 12 and 20, collagen expression increased between porous PDLLA particles and inside the particles (yellow arrows).

AestheFill® Journal & Study

1. Hyun MY, Lee Y, No YA, et al. Efficacy and safety of injection with poly-L-lactic acid compared with hyaluronic acid for correction of nasolabial fold: a randomized, evaluator-blinded, comparative study. Clinical and Experimental Dermatology. 2015;40(2):129-135.
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3. No YA, Seok J, Hyun MY, et al. Long-Term (24-Month) Safety Evaluation of Poly-DL-Lactic Acid Filler Injection for the Nasolabial Fold; A Multicenter, Open, Randomized, Evaluation-Blind, Active-Controlled Design. Plastic and Reconstructive Surgery. 2015;135(6):1074e-1075e
4. Chen SY, Lin JY, Lin CY. Letter to the editor regarding the article "Kwon T-R, Han SW, Yeo IK, et al. Biostimulatory effects of polydioxanone, poly-d, l lactic acid, and polycaprolactone fillers in mouse model. J Cosmet Dermatol. 2019;18(4):1002-1008.". Journal of Cosmetic Dermatology. 2020;19(4):1004-1005.
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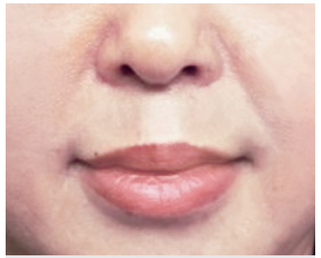
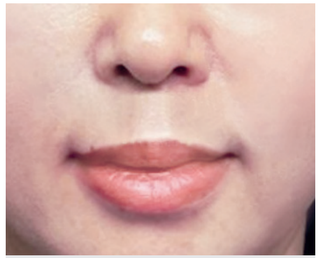
10. Lee SH, Ko K, Choo MS, et al. The effect and safety of polylactic acid and adipose-derived stromal vascular fraction cell as an injectable bulking agent in urologic field: A 24-week follow-up study. Journal of Biomedical Materials Research Part B, Applied Biomaterials. 2015;103(2):440-447.
11. Владлена АВЕРИНА, Наталья МАРКОВА, Светлана ПЕРВЫХ. ПОЛИМОЛОЧНАЯ КИСЛОТА: НОВЫЙ ВИТОК ПОПУЛЯРНОСТИ. Les Nouvelles Esth.tiques Украина. Нувель Эстетик 4 (122)/2020

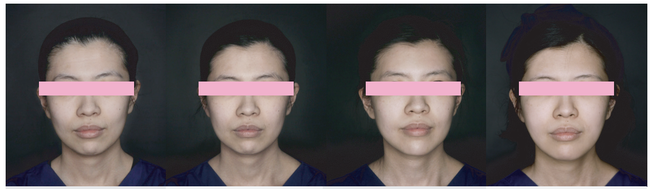
AestheFill® B&A



Before



After



Before Forehead, Tear trough Cheek | 3 Months later Chin, Cheek, Forehead | 12 Months later Temple, Nose | 24 Months later

What Happens As You Age

As you age, there is global volume loss of internal facial tissues due to progressive bone resorption, weakening of facial muscles, loss of collagen and deeper fat layer. Your skin loses its thickness and elasticity. As a result, there is a development of wrinkles, folds and skin laxity with overall saggy appearance. An analogy to the aging processes is very much like a balloon that is slowly deflating.

With AestheFill®, Time Can Be On Your Side.....

AestheFill® is an innovative treatment filler that continuously stimulating collagen formation, which helps to maintain a natural youthful appearance and correcting wrinkles due to aging. The process is gradual, and the result can last up to two years. If you want to look young in a natural way, AestheFill® is the ideal treatment for you.

AestheFill®
New Generation Facial Wrinkle Filler

REGEN Biotech, Inc.

AestheFill®

COLLAGEN REGENERATION NATURAL VOLUME

AestheFill® is an innovative treatment filler that continuously stimulating collagen formation, which helps to maintain a natural youthful appearance and correcting wrinkles due to aging.